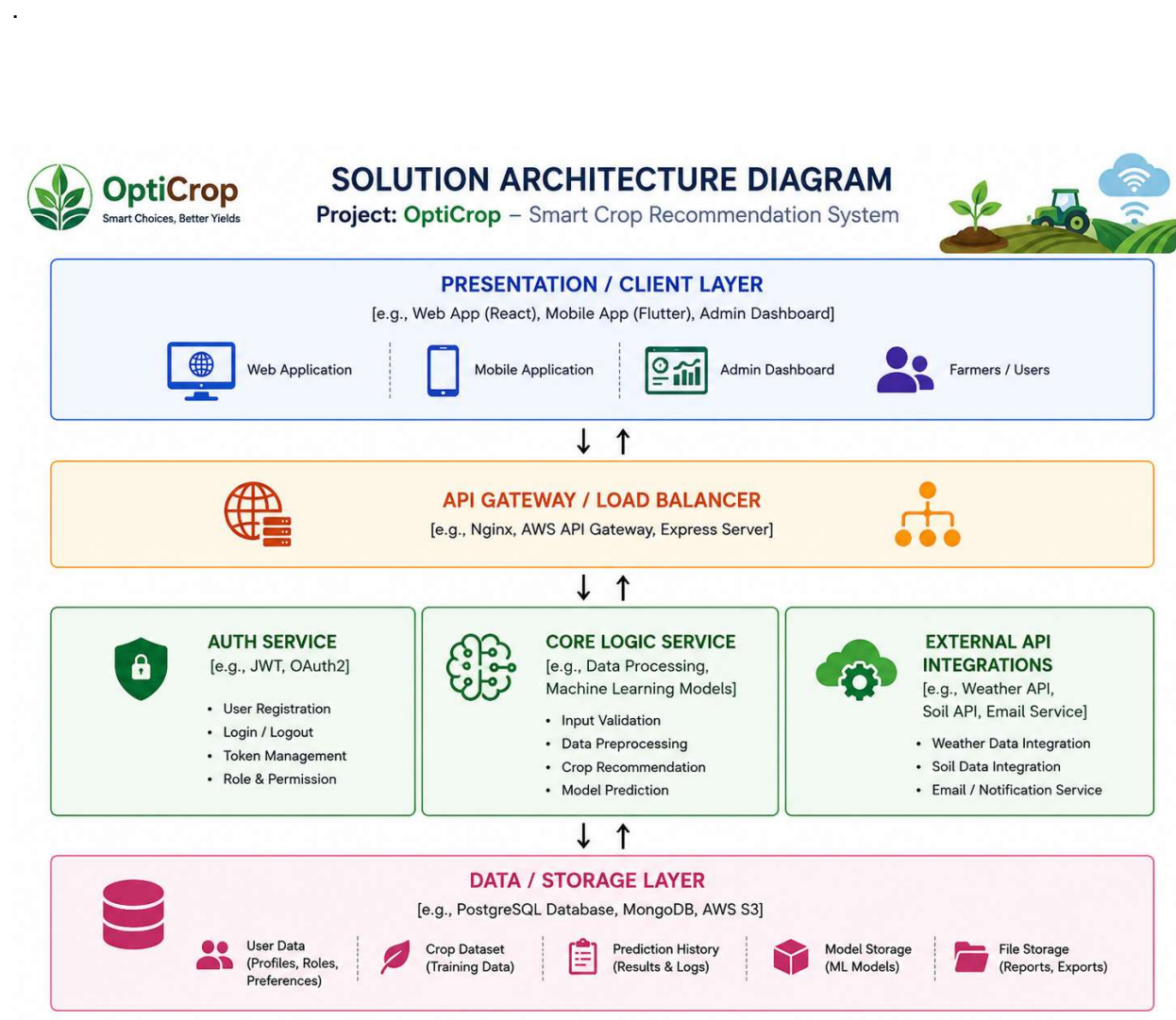


Solution Architecture

Date	28 June 2026
Team ID	
Project Name	Opti Crop Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	User interface where farmers enter soil and weather details and view the recommended crop.	HTML, CSS, Bootstrap, Flask (Frontend)
API Gateway	Receives user requests and forwards them to the prediction module. Handles communication between frontend and backend.	Flask
Core Logic Service	Processes user inputs, loads the trained machine learning model, and predicts the best crop.	<i>Python, Scikit-learn, Pandas, NumPy</i>
Database	Stores datasets, trained model files, and user input records (if applicable).	CSV Dataset, Pickle